

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 4

Aim: Performing Data Analysis using Calculations in Tableau

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

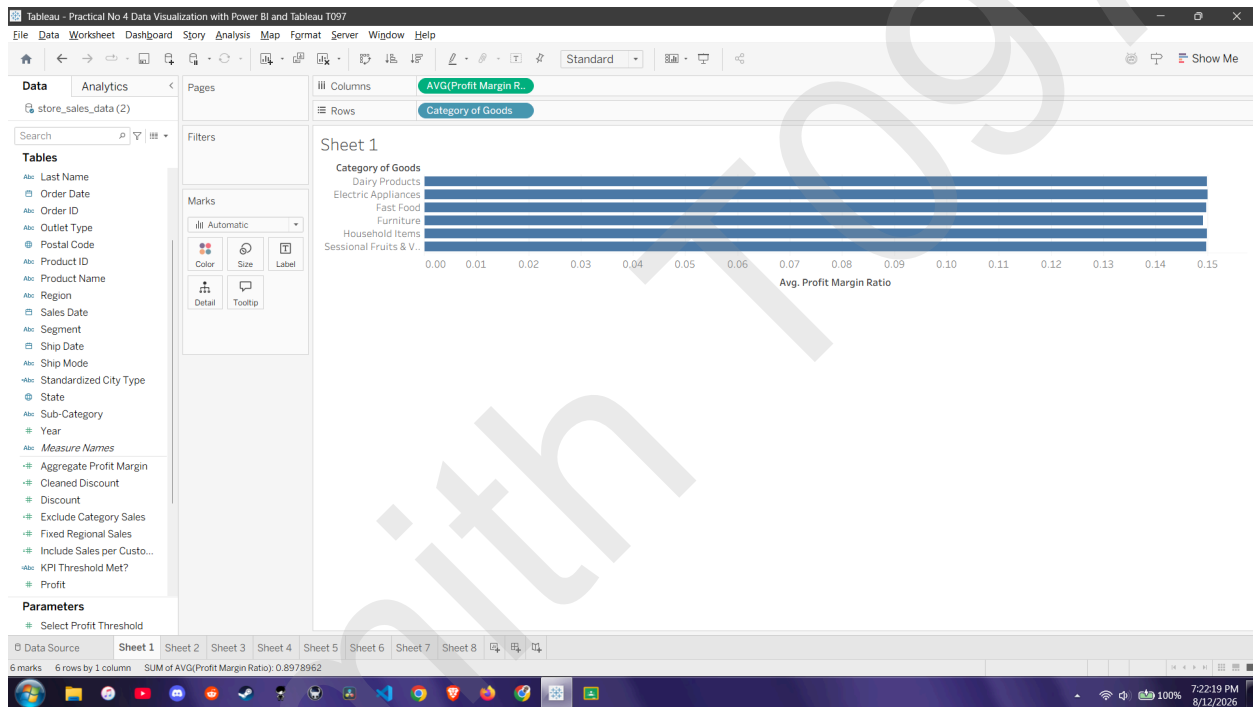
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a) Create basic calculated fields using formulas

Task 1: Basic Custom Calculation

- Create **Sheet 1**. In the Data Pane, click **Create Calculated Field** and name it Profit Margin Ratio.
- Formula: $[\text{Profit}] / [\text{Sales}]$. Click **OK**.
- Drag **Category of Goods** to the **Rows** shelf.
- Drag Profit Margin Ratio to the **Columns** shelf. Right-click the pill, hover over **Measure (Sum)**, and change it to **Average**.

After performing the following steps this is what you will be seeing:



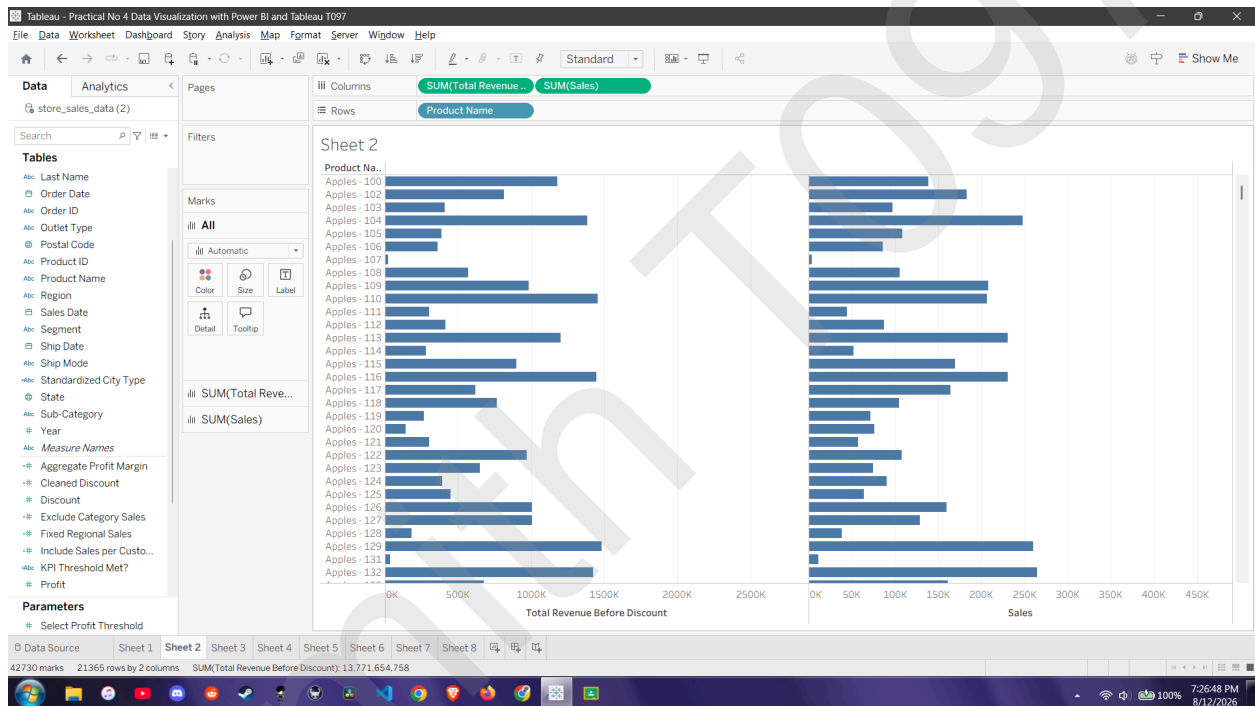
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b) Perform row-level calculations

Task 1: Row-Level Calculated Field

- Create **Sheet 2**. In the Data Pane, click **Create Calculated Field** and name it Total Revenue Before Discount.
- Formula: [Sales] * [Quantity]. Click **OK**.
- Drag **Product Name** to the **Rows** shelf.
- Drag Total Revenue Before Discount to the **Columns** shelf. Drag the raw **Sales** measure to the **Columns** shelf right next to it to visually compare the numbers.

After performing the following steps this is what you will be seeing:



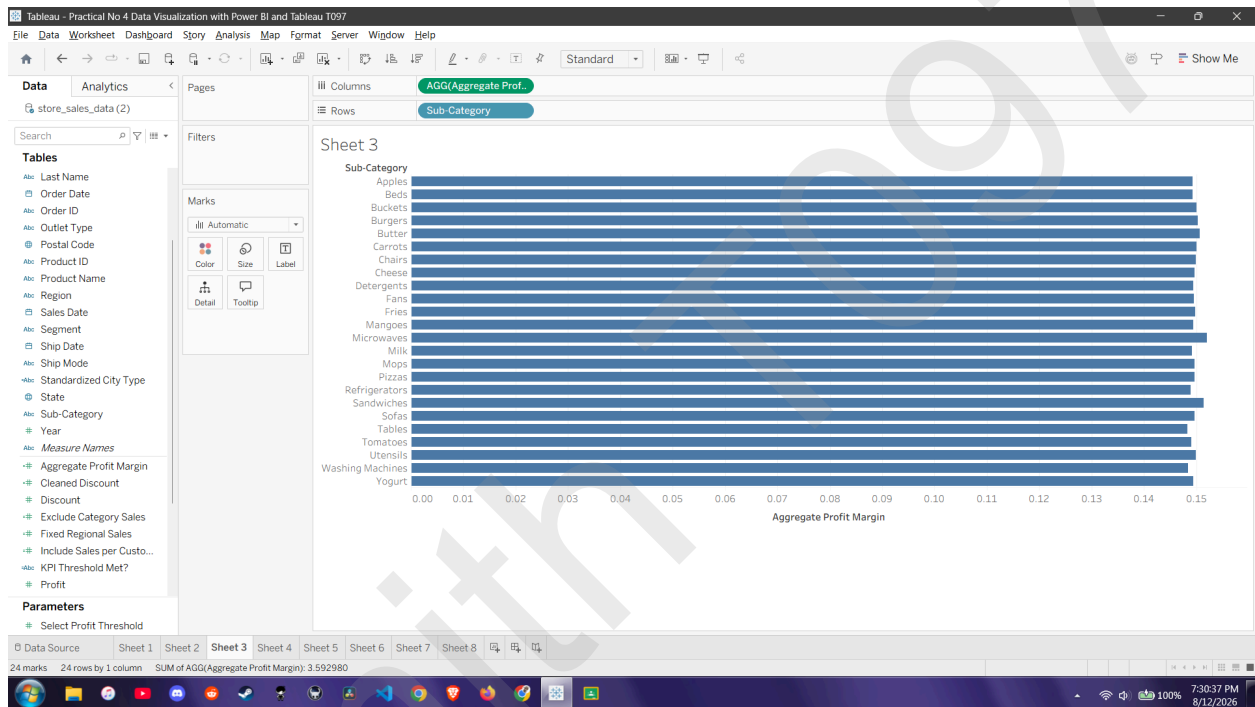
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c) Perform aggregate-level calculations

Task 1: Aggregate Calculated Field

- Create **Sheet 3**. In the Data Pane, click **Create Calculated Field** and name it Aggregate Profit Margin.
- Formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$. Click **OK**.
- Drag **Sub-Category** to the **Rows** shelf and Aggregate Profit Margin to the **Columns** shelf.

After performing the following steps this is what you will be seeing:



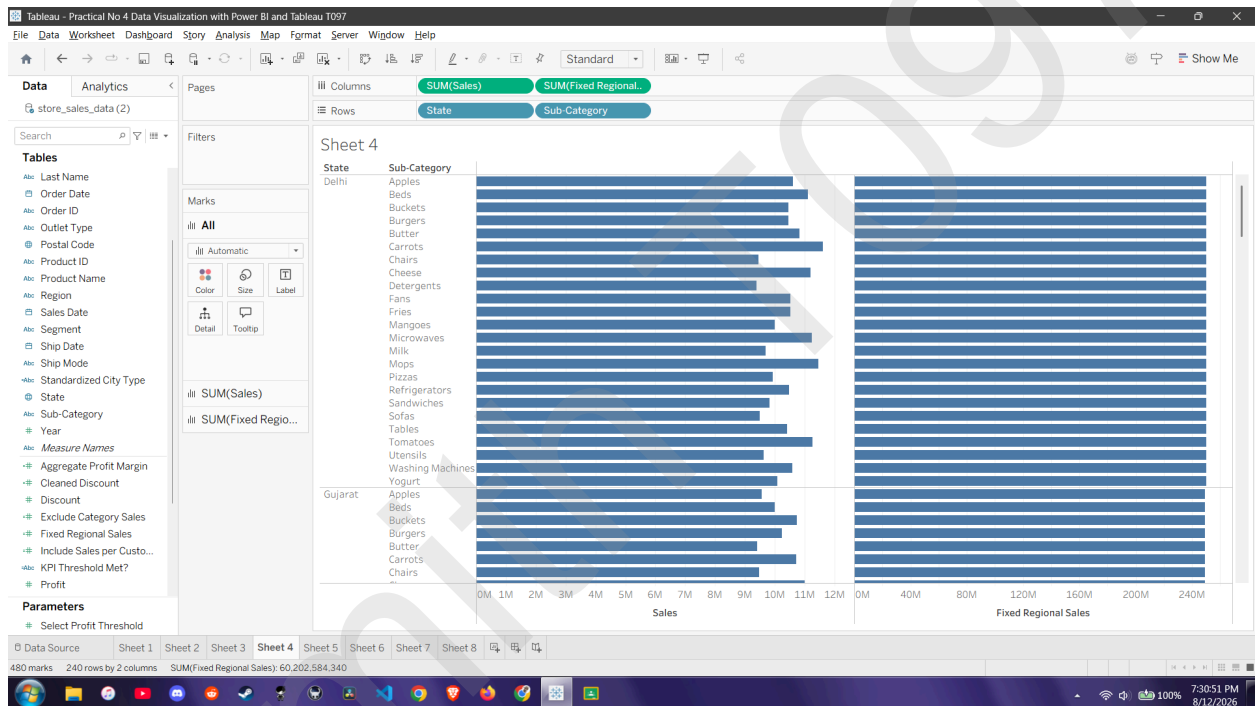
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d) Implement FIXED Level of Detail (LOD) calculations

Task 1: FIXED LOD Calculated Field

- Create **Sheet 4**. In the Data Pane, click **Create Calculated Field** and name it Fixed Regional Sales.
- Formula: { FIXED [State]: SUM([Sales]) }. Click **OK**.
- Drag **State** and **Sub-Category** to the **Rows** shelf.
- Drag **Sales** to the **Columns** shelf, then drag Fixed Regional Sales to the **Columns** shelf right next to it.

After performing the following steps this is what you will be seeing:



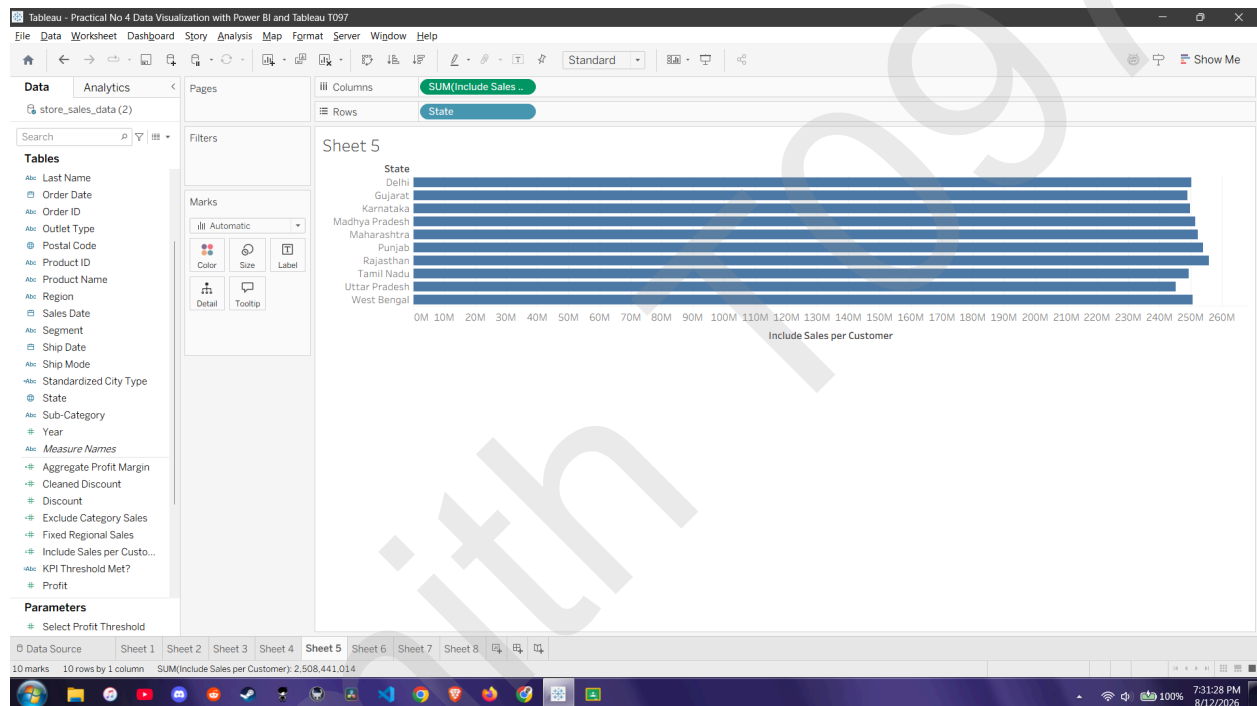
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e) Use INCLUDE and EXCLUDE LOD expressions

Task 1: INCLUDE LOD Calculated Field

- Create **Sheet 5**. In the Data Pane, click **Create Calculated Field** and name it Include Sales per Customer.
- Formula: { INCLUDE [Customer ID]: SUM([Sales]) }. Click **OK**.
- Drag **State** to the **Rows** shelf and Include Sales per Customer to the **Columns** shelf. Right-click the pill and change the Measure to **Average**.

After performing the following steps this is what you will be seeing:

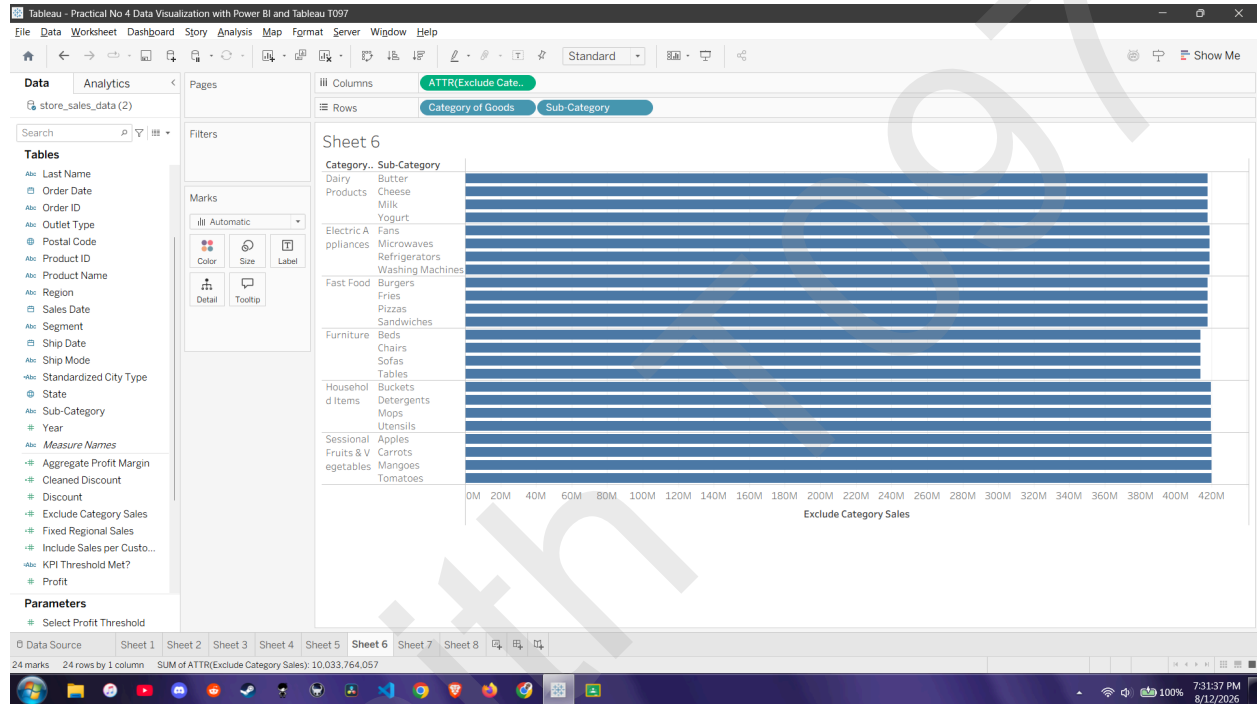


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Task 2: EXCLUDE LOD Calculated Field

- Create **Sheet 6**. In the Data Pane, click **Create Calculated Field** and name it Exclude Category Sales.
- Formula: { EXCLUDE [Sub-Category]: SUM([Sales]) }. Click **OK**.
- Drag **Category of Goods** and **Sub-Category** to the **Rows** shelf.
- Drag Exclude Category Sales to the **Columns** shelf.

After performing the following steps this is what you will be seeing:



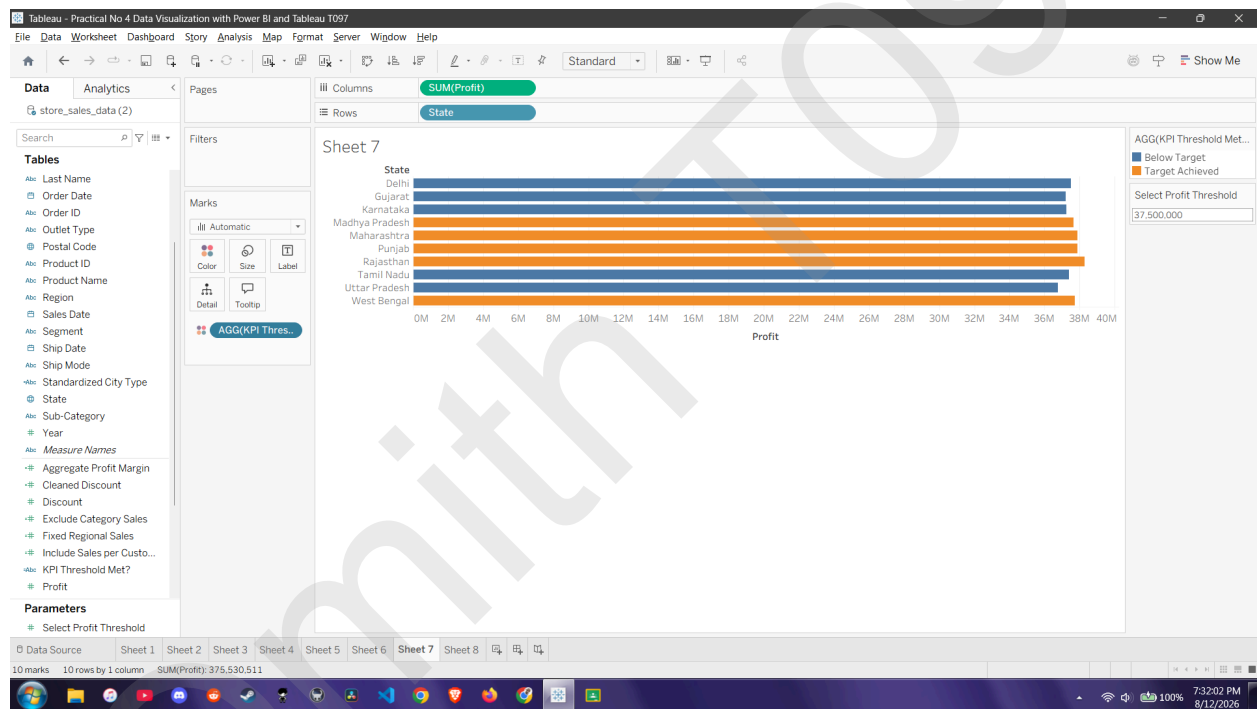
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f) Create and use parameters for dynamic analysis

Task 1: Interactive Parameter & Conditional Logic

- Create **Sheet 7**. In the Data Pane, select **Create Parameter**. Name it Select Profit Threshold. Set Data Type to **Float/Integer**, Current Value to **500**, and Allowable values to **All**. Click **OK**.
- Right-click the parameter in the Data Pane and choose **Show Parameter Control**.
- Create a Calculated Field named KPI Threshold Met?. Formula: IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved" ELSE "Below Target" END. Click **OK**.
- Drag **State** to the **Rows** shelf and **Profit** to the **Columns** shelf. Drag KPI Threshold Met? to the **Color** tile.

After performing the following steps this is what you will be seeing:



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g) Apply calculations to fix data issues and enhance insights

Task 1: Handling Nulls and Segmenting Data

- Create **Sheet 8**. Create a Calculated Field named Cleaned Discount. Formula: IFNULL([Discount], 0). Click **OK**.
- Create a Calculated Field named Standardized City Type. Formula: UPPER([City Type]). Click **OK**.
- Drag Standardized City Type to the **Rows** shelf and Cleaned Discount to the **Columns** shelf. Right-click the pill and change the Measure from Sum to **Average**.

After performing the following steps this is what you will be seeing:

